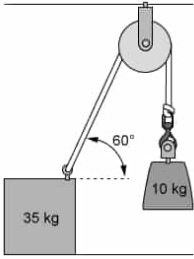


A 10 kg block is suspended in air by a single string that passes through a pulley and is attached on the other side to a 35 kg block that is on the verge of sliding on the ground.

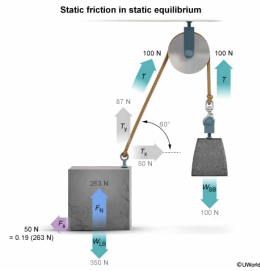


What is the coefficient of static friction between the larger block and the ground? (Note: The vertical component of the force the rope exerts on the large block is 87 N.)

- ☐ A. 0.11 [10%]  
☐ B. 0.15 [31%]  
☒ C. 0.19 [33%]  
☐ D. 0.23 [24%]

Correct  
 33% answered correctly

Explanation:



Attractive intermolecular forces between the atoms of two surfaces contribute to **frictional forces** that oppose motion along the interface between two objects that are **in direct contact** with one another. In **static equilibrium** (ie, motionless, with a balance of forces), a frictional force that resists the initiation of motion is called **static friction**. Static friction is oriented parallel to the contacting surfaces and arises in response to a force that attempts to move an object within the system.

The maximum magnitude of static friction  $F_s$  between two opposing surfaces is equal to the product of the **normal force**  $F_N$  and the **coefficient of static friction**  $\mu_s$ , a unitless coefficient related to the characteristics of the contacting surfaces:

$$F_s = \mu_s \cdot F_N$$

If the force applied to an object does not exceed  $F_s$ , the object will not move. Conversely, if  $F_s$  is **exceeded**, the object will accelerate and static friction becomes kinetic friction.

In this question, the two blocks and pulley are in static equilibrium. For the smaller block, the tension  $T$  in the rope is directed vertically upward (ie, the  $y$ -direction) and equal to the **weight**  $W_{SB}$  of the smaller block:

$$T = W_{SB} = mg$$

$$T = (10 \text{ kg}) \cdot \left(10 \frac{\text{m}}{\text{s}^2}\right) = 100 \text{ N}$$

On the left side of the pulley,  $T$  acts on the larger block and can be resolved into component forces oriented along the  $x$ - and  $y$ -axes. The horizontal component  $T_x$  equals:

$$T_x = T \cos \theta$$

$$T_x = (100 \text{ N}) \cdot \cos 60^\circ = 50 \text{ N}$$

and the vertical component  $T_y$  is:

$$T_y = T \sin \theta$$

$$T_y = (100 \text{ N}) \cdot \sin 60^\circ = 87 \text{ N}$$

Because the block is still in static equilibrium,  $F_s$  must be equal to  $T_x$ :

$$F_s = \mu_s \cdot F_N = 50 \text{ N}$$

Furthermore, in the  $y$ -direction the sum of  $F_N$  and  $T_y$  equals the weight of the larger block:

$$F_N + T_y = W_{LB}$$

Solving for  $F_N$  and substituting into the equation for  $F_s$  gives:

$$\mu_s \cdot (W_{LB} - T_y) = 50 \text{ N}$$

$$\mu_s \cdot \left[ \left(35 \text{ kg} \cdot 10 \frac{\text{m}}{\text{s}^2}\right) - 87 \text{ N} \right] = 50 \text{ N}$$

$$\mu_s \cdot (263 \text{ N}) = 50 \text{ N}$$

$$\mu_s = \frac{50 \text{ N}}{263 \text{ N}} \approx 0.19$$

Therefore,  $\mu_s$  between the large block and the ground is 0.19 (**Choice C**).

**Educational objective:**

Static friction is an opposing force within a system in static equilibrium that acts to prevent motion along the interface between two objects in direct contact. The maximum magnitude of static friction is the product of the normal force and the coefficient of static friction.

Time spent: 0 sec  
 Subject:  
 Physics

QID: 402705  
 System:  
 Mechanics and Energy

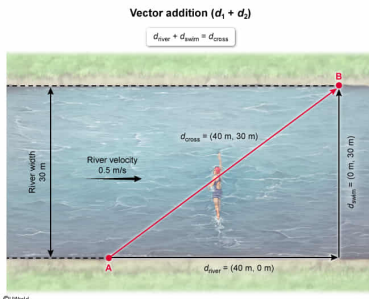
A swimmer enters a straight river at Point A and attempts to swim directly across to the opposite shore. However, the downstream velocity of the river is 0.5 m/s, which causes the swimmer to arrive at Point B on the opposite shore after 80 seconds of swimming. If the river is 30 m wide, what is the minimum diagonal distance between Points A and B? (Note: Assume the swimmer does not act to oppose the flow of the river.)

- ☐ A. 30 m [4%]  
☐ B. 40 m [32%]  
☒ C. 50 m [59%]  
☐ D. 60 m [3%]

Correct

60% answered correctly

#### Explanation:



**Scalar quantities** refer to physical properties that are characterized by only size or magnitude (eg, time, mass). Conversely, **vector quantities** are characterized by both a **magnitude and direction** (eg, velocity). Therefore, a vector, such as displacement  $d$ , oriented on an  $xy$ -coordinate system is used when expressing a vector quantity within two-dimensional space:

$$d = (x, y)$$

Due to this shared convention, the  $x$ - and  $y$ -components of two vectors  $d_1$  and  $d_2$  may be **added** together such that the coordinates of the resulting vector  $d_R$  are expressed as:

$$d_R = d_1 + d_2$$

$$d_R = ([x_1 + x_2], [y_1 + y_2])$$

Furthermore, the magnitude of any vector  $|d|$  is equal to the square root of the sum of the  $x$ - and  $y$ -components squared:

$$|d| = \sqrt{x^2 + y^2}$$

If Point A along the river's shore is set as the origin, the width of the river can be oriented along the  $y$ -axis of an  $xy$ -coordinate system. Consequently, the vector corresponding to the swimmer's effortful crossing  $d_{\text{swim}}$  depends only on the width of the river:

$$d_{\text{swim}} = (0 \text{ m}, 30 \text{ m})$$

The flow of the river acts to passively transport the swimmer down the river in the horizontal ( $x$ -axis) direction. As such, the swimmer's motion due to the river flow  $d_{\text{river}}$  may be expressed in terms of the velocity of river flow multiplied by the time it takes the swimmer to cross the river:

$$d_{\text{river}} = \left( \left[ 0.5 \frac{\text{m}}{\text{s}} \cdot 80 \text{ s} \right], 0 \text{ m} \right)$$

$$d_{\text{river}} = (40 \text{ m}, 0 \text{ m})$$

Consequently, the vector corresponding to the total motion of the swimmer  $d_{\text{cross}}$  is the sum of the components of the swimmer's motion along both the  $x$ - and  $y$ -axes:

$$d_{\text{cross}} = d_{\text{swim}} + d_{\text{river}}$$

$$d_{\text{cross}} = ([0 \text{ m} + 40 \text{ m}], [30 \text{ m} + 0 \text{ m}]) = (40 \text{ m}, 30 \text{ m})$$

Furthermore,  $|d_{\text{cross}}|$  is equal to:

$$|d_{\text{cross}}| = \sqrt{(40 \text{ m})^2 + (30 \text{ m})^2}$$

$$|d_{\text{cross}}| = \sqrt{1600 \text{ m}^2 + 900 \text{ m}^2} = 50 \text{ m}$$

Therefore, the distance between Points A and B on opposites sides of the river is at least 50 m (**Choice C**).

**Educational objective:**

Vectors are used to express physical quantities characterized by both a magnitude and a direction. Vectors may be added to one another such that a resultant vector is formed with unique magnitude and direction.

**Time spent:**0 sec  
**Subject:**  
Physics

**QID:**402706  
**System:**  
Mechanics and Energy